VENTUNO MOTO CIRCUIT | CONNECTION SCHEMATIC

Arduino VENTUNO Q / App Lab on a laptop / optional physical Knob control

1 DC BARREL POWER SUPPLY

Use an Arduino-recommended supply.
Match the board voltage, polarity and power requirements.

DC POWER

2 USB WEBCAM

Point it at the player's hands.
USB camera capture runs on the board.
Logitech BRIO was used in this demo.

USB CABLE

3 MODULINO KNOB (OPTIONAL)

Connect using a keyed Qwiic cable.
MCU sketch reads rotation and press.
Gears are automatic in simplified mode.
Knob press can start / pause the game.

QWIIC CABLE

DC BARREL INPUT

ARDUINO

VENTUNO Q

BOARD-SIDE PROCESSING

Camera capture + local hand tracking
Palm and landmark models: QNN NPU
Steering / wrist geometry: CPU
App Lab gateway: port 7000

USB-A HOST PORT

USB-C DATA / DEVICE

MCU QWIIC / Wire1 (I2C)

MCU input reaches Linux via Router Bridge.

USB DATA CABLE
Board USB-C to laptop USB

4 LAPTOP / PC

Arduino App Lab runs / manages the app.
Browser renders the 3D racing scene.
Laptop GPU handles WebGL graphics.
Open http://localhost:7000/
USB setup forwards the board web port.
Optional engine audio: laptop speakers.

Display cable

OPTIONAL EXTERNAL DISPLAY

Connect to the laptop's display output.
HDMI / USB-C depends on the laptop.

CONNECT IN THIS ORDER

1. Power VENTUNO Q from its barrel input before connecting the laptop USB cable.
2. Plug the webcam into board USB-A host; connect the optional Knob to the MCU Qwiic bus.
3. Connect board USB-C data to laptop, run the imported App Lab project, then open the browser game.

Connector positions are illustrative, not a physical port map. No GPIO wiring, actuator or board audio speaker is required.

Connection reference: project README and sketch/sketch.ino (Modulino.begin(Wire1)); power/setup: docs.arduino.cc/tutorials/ventuno-q/first-setup/